

1. Benefit of DL Frameworks:

- A. by using DL frameworks, which are different kinds of specific libraries such as TensorFlow and PyTorch, we don't need to write low-level detail from scratch (such as mathematical implementations of forward pass, backward pass, and weight adjustment in backpropagation). this can significantly improve productivity in academic research.
- B. to avoid implement low-level detail from scratch, DL frameworks provide a number of functions ($\equiv$ APIs) enable you to work at the highest abstraction level.
for example, with functions in Keras, a stand alone library which can be imported from multiple DL frameworks, we can use "individual layer in a neural network" as basic building block, rather than "individual neuron in a neural network".